

ECC Binomial Model

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Chapter 1

The One-Period Binomial Model for European Contingent Claims

1.1 The Market

Definition 1. The initial *market parameters* are defined by

- $S_0 > 0$, the initial price of the risky asset;
- $u > 0$, the up factor;
- $d > 0$, the down factor;
- $r > -1$ such that the risk-free asset grows by a factor of $1 + r$ at each time step.

Definition 2. The sample space for the ECC binomial model is $\Omega = \{u, d\}$, representing a single up or down move. We apply the probability measure P such that $P(\{u\}) > 0$ and $P(\{d\}) > 0$.

Definition 3. The filtration for the ECC binomial model is the natural filtration, defined as $\mathcal{F}_0^0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_1^0 = 2^\Omega$.

1.2 Asset Prices

Definition 4. The *bond* is the risk-free asset, with price process defined by $B_0 = 1$ and $B_1 = 1 + r$.

Definition 5. The *stock* is the risky asset, with price process defined by S_0 at time 0 and

$$S_1(\omega) = \begin{cases} uS_0 & \text{if } \omega = u, \\ dS_0 & \text{if } \omega = d. \end{cases}$$

Definition 6. A *trading strategy* is a pair of $\mathcal{F}^0$ -adapted processes $\varphi = (\alpha, \beta)$, where α_t represents the number of shares of the stock held at time t and β_t represents the number of shares of the bond held at time t . The value of the trading strategy at time t is given by $V_t(\varphi) = \alpha_t S_t + \beta_t B_t$.

1.3 Arbitrage

Definition 7. There exists *arbitrage* or equivalently an *arbitrage opportunity* if there exists a trading strategy $\varphi = (\alpha, \beta)$ such that

- $V_0(\varphi) = 0$;
- $V_1(\varphi) \geq 0$;
- $E[V_1(\varphi)] > 0$.

Definition 8. The market is *arbitrage-free* if there are no arbitrage opportunities.

Definition 9. A *risk-neutral probability* is a probability measure q on Ω such that

$$q = \frac{(1+r) - d}{u - d}$$

Lemma 10. *The market is arbitrage-free if and only if $d < 1 + r < u$.*

Proof. • $(\Rightarrow)$ Assume the market is arbitrage-free. If $(1+r) \leq d$, then define the trading strategy $\varphi = (\alpha, \beta)$ by $\alpha_0 = 1, \beta_0 = -S_0$. Then $V_0(\varphi) = 0$ and $V_1(\varphi) = S_1 - S_0(1+r) \geq 0$ with positive probability. Thus, $E[V_1(\varphi)] > 0$, which contradicts the assumption of no arbitrage. Similarly, if $(1+r) \geq u$, then define the trading strategy $\varphi = (\alpha, \beta)$ by $\alpha_0 = -1, \beta_0 = S_0$. Then $V_0(\varphi) = 0$ and $V_1(\varphi) = -S_1 + S_0(1+r) \geq 0$ with positive probability, leading to a similar contradiction.

- $(\Leftarrow)$ Assume $d < 1 + r < u$. Let $\varphi = (\alpha, \beta)$ be any trading strategy such that $V_0(\varphi) = 0$. Then $\beta_0 = -\alpha_0 S_0$ and

$$V_1(\varphi) = \begin{cases} \alpha_0 S_1 + \beta_0 S_0(1+r) = \alpha_0 S_0(u - (1+r)) & \text{if } \omega = u, \\ \alpha_0 S_1 + \beta_0 S_0(1+r) = \alpha_0 S_0(d - (1+r)) & \text{if } \omega = d. \end{cases}$$

Since $d < 1 + r < u$, we note that $V_1(\varphi) < 0$ with positive probability, so the market is arbitrage-free. □

Lemma 11. *The risk-neutral probability q is in the unit interval $(0, 1)$ if and only if $d < 1 + r < u$.*

Proof. Follows directly from the definition of q and the condition $d < 1 + r < u$. □

Definition 12. The *risk-neutral measure* Q on $\Omega = \{u, d\}$ is defined by $Q(\{u\}) = q$ and $Q(\{d\}) = 1 - q$.

Lemma 13. *Q is a probability measure on $(\Omega, \mathcal{F})$.*

Proof. We have $Q(\{u\}) + Q(\{d\}) = q + (1 - q) = 1$ and both weights are in $(0, 1)$ by 11. □

Lemma 14. *The discounted stock price satisfies*

$$E^Q \left[\frac{S_1}{1+r} \right] = S_0.$$

Proof. Direct computation: $E^Q[S_1/(1+r)] = q \cdot uS_0/(1+r) + (1-q) \cdot dS_0/(1+r) = S_0(qu + (1-q)d)/(1+r) = S_0 \cdot (1+r)/(1+r) = S_0$. □

1.4 European Contingent Claims

Definition 15. A *European Contingent Claim* (ECC) is a random variable $\Phi : \Omega \rightarrow \mathbb{R}$. We write $\Phi_u = \Phi(u)$ and $\Phi_d = \Phi(d)$ for its two possible payoffs.

Example 1.1. The European call option with strike $K > 0$ has payoffs $\Phi_u = \max(uS_0 - K, 0)$ and $\Phi_d = \max(dS_0 - K, 0)$.

Example 1.2. The European put option with strike $K > 0$ has payoffs $\Phi_u = \max(K - uS_0, 0)$ and $\Phi_d = \max(K - dS_0, 0)$.

1.5 Replicating Strategies

Definition 16. A trading strategy $\varphi = \alpha, \beta$ is a *replicating strategy* for an ECC Φ if $V_1(\varphi) = \Phi$.

Theorem 17. Under the arbitrage-free condition, every ECC Φ has a replicating strategy.

Proof. The replication conditions form a linear system in (α, β) whose coefficient matrix has determinant $(u - d)S_0(1 + r) \neq 0$ by 10. Solving by substitution yields the stated formulas. $\square$

1.6 Pricing

Theorem 18. The initial value of the unique replicating strategy of an ECC Φ is

$$V_0 = \frac{1}{1 + r} E^Q[\Phi] = \frac{q\Phi_u + (1 - q)\Phi_d}{1 + r}.$$

Proof. Substituting the formulas from 17 into $V_0 = \alpha S_0 + \beta$ and simplifying using $q = ((1 + r) - d)/(u - d)$ yields $V_0 = (q\Phi_u + (1 - q)\Phi_d)/(1 + r) = E^Q[\Phi]/(1 + r)$. $\square$

Corollary 19. The arbitrage-free price of a European call option with strike K is

$$V_0 = \frac{q \max(uS_0 - K, 0) + (1 - q) \max(dS_0 - K, 0)}{1 + r}.$$

Corollary 20. The prices of a European call C_0 and put P_0 with the same strike K satisfy put-call parity:

$$C_0 - P_0 = S_0 - \frac{K}{1 + r}.$$